

Ordinal Innovation: The Prime Signal from Succession Order Alone

The circularity weakened — the extraction no longer references the divisor lattice

Alexander Pisani & Claude (Anthropic) · June 2026 · companion note · verified by
 ordinal_innovation.py

Framing. Every prior extraction of the prime signal imported the divisor lattice: Stage 1’s $\mathbb{Z}/g$ sub-period test, the gateway’s clean-split pattern, the Möbius inversion’s divisor sums, and the divisor-conditioned innovation of the previous experiment. This experiment replaces the conditioning set with one defined purely by **succession order** — all predecessors — and the prime signal survives intact. Claims graded Fact / Reading / Lead.

1. The result

$$H(a \bmod n \mid \{a \bmod k : 2 \leq k \leq n-1\}) = \Lambda(n),$$

verified by exact enumeration for $n = 2..18$ to 6.2×10^{-15} . The procedure references only: the succession-cycle family (Stage 1’s clocks, which succession builds natively), the ordinal sweep (condition on everything *before* n), the shared stream (one a read by every clock), and Shannon conditional entropy. **Divisor knowledge appears nowhere in the extraction.** (*Fact.*)

Three corollaries, all verified:

- **A prime is a channel whose innovation is its entire content.** The novelty ratio $\text{innovation}/\ln n$ equals 1 exactly iff n is prime; a prime power p^k innovates by fraction $1/k$; a composite innovates zero. Primality is the *pure-novelty* condition of an order-indexed sweep. (*Fact.*)
- **Cumulative innovation is the second Chebyshev function.** $\sum_{m \leq n} \text{innovation}(m) = \ln \text{lcm}(1..n) = \psi(n)$ — the total information of the ordinal channel bank is the central object of prime distribution. (*The identity $\ln \text{lcm}(1..n) = \psi(n)$, and $\psi(n) - \psi(n-1) = \Lambda(n)$: classical, flagged. The realisation as the bank’s accumulated innovation: Reading.*)
- **The decay profile is the factorisation, read by order.** Sweeping the conditioning set $I_m(n) = H(a \bmod n \mid \text{channels } 2..m)$: a prime’s profile stays *flat* at $\ln n$ through the entire sweep (nothing before it explains any of it — $n = 13$: 2.565 at every m); a composite’s steps down as the sweep covers each prime-power part ($n = 12$: $2.485 \rightarrow 1.792 \rightarrow 0.693 \rightarrow 0$; $n = 16$: steps exactly at $m = 2, 4, 8$). (*Fact.*)

2. What this does to the standing circularity

The circularity is **weakened, precisely**: the divisor lattice is no longer an *input to the extraction procedure*. The procedure is succession-native — an order sweep plus an entropy — and primality emerges as its output (the channels of novelty ratio 1). What the move does *not* do, stated with

equal precision: the channel family itself is still indexed by the integers (the Stage 1 relabel stands at the family level — the girths are the integers, and succession’s cycles come labelled by their return times); the uniform superposition and the full readout are still assumed; and the divisor structure has not vanished — it has moved *out of the procedure and into the data*, where it lives in the joint distribution of the readouts (the CRT). So the relocation of §19 is advanced one genuine step: from “the extraction needs the divisor lattice” to “the extraction needs only order; the lattice announces itself in what the measurements say.” (*Reading; the relabel caution stands.*)

3. The prime number theorem as an information rate (flagged Lead, no leverage)

Since the bank’s cumulative innovation is $\psi(n)$, the prime number theorem $\psi(n) \sim n$ reads, in the framework’s variables: **the ordinal channel bank gains asymptotically one nat of innovation per step of succession** — counting, read through all its cycles, produces new information at unit rate. The fluctuation of that rate around linearity is the classical domain of the deepest open questions of prime distribution, including the Riemann Hypothesis, and **no leverage on any of them is claimed or implied**; the reading is picture-deepening only, in exactly the sense of the paper’s §23 discipline. (*PNT: classical. The information-rate reading: Reading. Anything beyond: not claimed.*)

4. Honesty boundary

The underlying identities are textbook ($\psi(n) = \ln \text{lcm}(1..n)$; $\Lambda(n) = \psi(n) - \psi(n-1)$; CRT). The contribution is the demonstration that the prime-signal extraction can be formulated with an order-defined conditioning set — succession-native, divisor-free in procedure — together with the pure-novelty characterisation of primes, the decay-profile factorisation signature, and the information-rate reading of ψ . The relabel caution travels; no arithmetic question is claimed resolved.

The originating question — whether the innovation formulation could weaken the standing circularity — arose in session with Alexander Pisani; the experiment and this note were developed jointly with Claude (Anthropic). June 2026.